TrustCore

Real-time trust scoring for payments, identity documents, and access

Every event gets a score, a decision, and a reason, as it happens.

IBM Z Datathon 2026 | Track: Real-time AI for Critical Decisions

The problem: decisions come too late

Fraud is caught after the money moves

A fraudulent transfer clears, a fake ID opens an account, an insider copies files. Batch checks and manual review notice afterwards.

Threats are getting harder to spot

AI-generated phishing, synthetic identities, and forged documents outpace slow tools. Each risk has its own separate system.

Who gets hurt most

A delivery worker paid with a fake UPI screenshot. A small shop owner who onboards a fraudulent vendor. A gig platform flooded with fake accounts. None of them has a fraud team or time to investigate.

The solution: one engine, three decisions

Payments

Flags unusual amounts, hours, cities, and rapid bursts

Identity documents

Catches expiry before issue, low face match, forged formatting

Access events

Spots off-hours logins, bulk file access, new devices

Every event returns:

a trust score from 0 to 100 | a flag: trusted, review, or high-risk | a plain-English reason

Clear-cut events pass or fail automatically. Humans review only the borderline cases.

How it works

Event
payment, ID document, or login

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Scoring engine
rules + Isolation Forest

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Flask API
score on demand

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Dashboard
review and filter

Two signals, one score

Rules catch known red flags, such as an amount 10x the user's norm. Isolation Forest learns normal behavior and flags the unexpected.

Score = 60% ML + 40% rules

Why it is explainable

Every flagged event lists which rules fired. A reviewer sees why, not just a number. Flags: 70+ trusted, 40-69 review, below 40 high-risk.

Live demo: same engine, opposite outcomes

Suspicious payment

Rs 25,000 vs usual Rs 800. 3 AM. Kolkata, home city Lucknow. 6 payments in the last hour.

Trust score 8.3

HIGH RISK, with 4 reasons listed

Normal payment

Rs 500 vs usual Rs 450. 2 PM. Home city. 1 payment in the last hour.

Trust score 98.9

TRUSTED, no anomalies

On simulated data, all 89 planted frauds were flagged. This is a pipeline sanity check, not a real-world accuracy claim.

IBM Z: built for in-transaction decisions

Today

Prototype built and tested on a laptop with open-source Python (scikit-learn, Flask). It does not run on IBM Z yet.

Planned integration

1. Train off-platform, deploy for inference through Machine Learning for z/OS.
2. Run the scoring API as a Linux workload on LinuxONE.
3. Add a small neural scorer to use the Telum on-chip AI accelerator.

Why it fits: the trust decision must happen before the transaction completes, with data staying on the platform.

We will validate latency on IBM Z before claiming any performance.

Impact and next steps

Who benefits

- Gig workers and small merchants
- Small businesses onboarding users
- Security and compliance teams
- Auditors and regulators

One engine replaces three siloed tools, with a reason for every decision to support audits and rules like DPDP and GDPR.

Next steps

- Benchmark on public fraud data
- Report precision and false positives
- Measure scoring latency
- Bring identity and access modules to the live dashboard
- Deploy and test on IBM Z